

Paper:

Wide-Range and Real-Time 3D Tracking of a Moving Object by Dynamic Compensation Based on High-Speed Visual Feedback

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To apply dynamic compensation in increasing productivity and realizing high-speed manipulation and manufacturing, a wide-range 3D tracking system combining a high-speed 3D compensation module (also called a delta robot because of its structure) and a robotic arm is developed. During tracking, the cameras collect 1000 images per second, and the tracked object position in the real-world robotic-arm base coordinate system is calculated using the collected pixel data and camera calibration parameters. Corresponding commands are generated for the robotic arm and delta robot to maintain their positions relative to the tracked object. To generate these commands, encoder feedback from the robotic arm and delta robot is utilized. The cameras provide 1000-Hz visual feedback, and the high-speed compensation module provides 1000-Hz response to movement with high accuracy. The robotic arm achieves this accuracy over a wide range. Successful tracking along the xy -axis is achieved. Even when the object moves faster, the tracking accuracy is high along the x - and z -axes. This study demonstrates the feasibility of using a delta robot to achieve wide-range object tracking, which may contribute to higher-speed manipulation of robotic arms to increase productivity.

Keywords: high-speed vision, 3D tracking, dynamic compensation, robot control, 3D measurement

1. Introduction

1.1. Collaborative Robots

Since societal demand is becoming increasingly specific and varies in different aspects, industrial production is changing accordingly. An increasing number of companies are shifting from large-scale production of a few va-

rieties of products to small-scale production of many varieties of products based on customer-specific demands. With this trend, the ability of industrial robots to manage different types of manufacturing and their intelligence in receiving different types of commands are becoming increasingly important [1]. Therefore, in contrast to past industrial robots aimed at mass production, new collaborative robots have been invented and produced for complicated or resilient manufacturing.

The concept of collaborative robots is working with humans. Humans have the highest levels of resilience and intelligence, with the brain being the most complicated organ. Working with human workers who can accomplish various types of complicated tasks, collaborative robots can generally “learn” these features. Robotic engineering attempts to improve collaborative robots to learn human actions and purposes.

However, most collaborative robots continue to operate separately from or sequentially with humans. One of the main reasons for this is that the action speed and acceleration of collaborative robots are restricted for safety. Collaborative robots generally require a function to stop automatically when contact with humans or obstacles is detected. Therefore, their speed and acceleration are typically limited to stopping as soon as possible during an emergency. With low-level action speed and acceleration, the accuracy of collaborative robots during movement is low; thus, dynamic compensation is required.

1.2. Dynamic Compensation

Dynamic compensation is a fundamental and sophisticated approach in the field of control systems that aims to address the challenges posed by dynamic disturbances in diverse engineering applications. Dynamic compensation involves the application of advanced strategies and techniques to counteract the effects of dynamic disturbances and ensure stability and optimal performance of complex

systems. This concept has gained prominence as industries demand higher precision, efficiency, and adaptability in the face of dynamic environmental changes.

One type of dynamic compensation involves the use of an individual dynamic compensation system, in which the dynamic compensation module directly provides position compensation based on real-time feedback and system observations. Similar concepts, such as macro-micro configurations of robotic arms, have been studied in detail [2–5]. This idea splits a large system into two complete subsystems from the hardware and software aspects to improve positioning accuracy. This adaptability enables the system to respond to changing conditions and disturbances, offering a robust solution for applications in which uncertainties are prevalent.

Using this add-on module concept, a dynamic compensation approach was developed for improving the autonomy of industrial robots with real-time positioning abilities [6–9]. The intelligence of a system is divided into two parts: intelligence of action for motion control and intelligence of planning for motion and task planning. These two aspects of intelligence belong to two subsystems. Action-level intelligence represents a low-level intelligence architecture and is referred to as adaptability for motion control without sacrificing motion speed and absolute accuracy simultaneously. Real-time adaptation to both system and environmental uncertainties enables a robot to focus on implementing high-level intelligence. Ideally, action-level intelligence is realized with a high-frequency update rate to compensate for the high-frequency portion of online uncertainties and changes, whereas planning-level intelligence can be implemented with a low-frequency update rate to handle the low-frequency portion of the uncertainties. Traditional industrial robots are designed to perform coarse global motions by focusing on planning-level intelligence [9]. Concurrently, an add-on robotic module with high-speed actuators and sensory feedback is controlled to perform accurate local compensation by implementing action-level intelligence. The proposed approach is based on the following assumptions.

1. The systematic uncertainty of a typical multi-joint industrial robot, along with environmental uncertainty, contributes to the accumulated positioning error denoted by δ , which is assumed to be bounded by Σ as $|\delta| \leq \Sigma$ ($\Sigma > 0$).
2. With high-speed sensors locally installed to observe a manipulation tool point and the manipulated target, the accumulated error, δ , is observed and estimated as the relative error $\hat{\delta}$ in the sensory feature space.
3. A high-speed robotic module with a working range broader than the bound Σ in a Cartesian space achieves high-speed, accurate compensation with real-time feedback. Ideally, a robotic module should have a higher frequency update rate and higher accountability than that of a typical industrial robot [10].

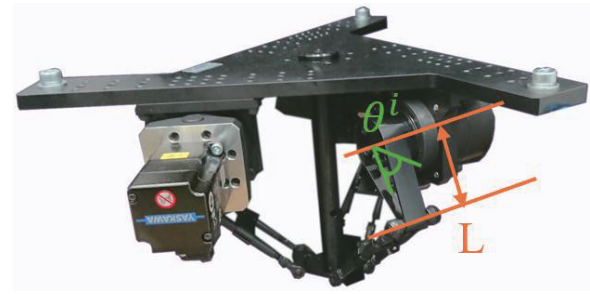


Fig. 1. Overview of the high-speed three-dimensional (3D) compensation module, with the common motor arm length, L , and the motor angles θ^i , $i = 1, 2, 3$, for the three motors assigned with number [1].

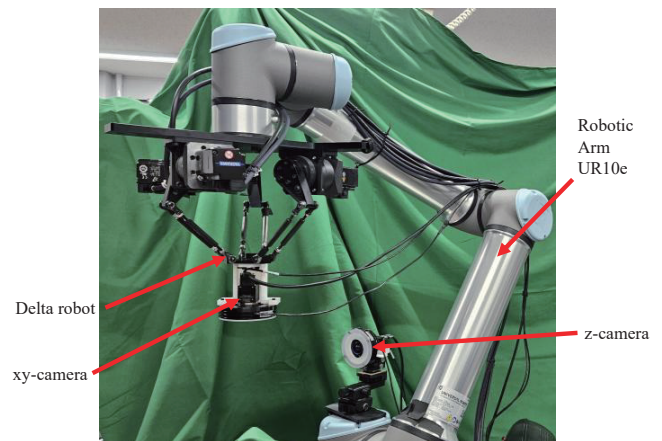


Fig. 2. A real-world overview of the 3D tracking system.

1.3. Previous Study

Murakami et al. developed a dynamic compensation module with high-speed visual feedback. This compensation module was a three-axis delta robot, as shown in **Fig. 1**, designed to perform highly accurate movements at high acceleration with motors that are responsible for high-frequency inputs [1]. This robot can maintain the orientation of moving stage, with a maximum movement range of approximately ± 110 mm along the x - and y -axes and 0–180 mm along the z -axis.

This high-speed three-dimensional (3D) compensation module was installed under a commercial collaborative industrial robot (UR10e, Universal Robots), as shown in **Fig. 2**, to perform automated bead art assembly [10]. A picking tool configured with a high-speed camera was developed and attached to the moving endpoint of the compensation module. Plastic balls with diameters of 5 mm and various colors were placed on an electric turntable, which made the balls move randomly to simulate uncertainties. When the controller received an ordered bead-art configuration, the assembly robot and high-speed compensation add-on module were required to provide accurate tracking of the desired color balls using high-speed visual feedback, lowering the height of the assembly robot tool center. The picking tool on the module endpoint picked the ball when the height of the endpoint was suitable for pick-

ing. Finally, the main assembly robotic arm brought the ball to the setting board to build the bead art. This process was repeated several times until the art was completed.

This study proved the feasibility of a dynamic compensation approach for real-time accurate positioning using the developed high-speed 3D compensation module and an algorithm for image processing under high-speed vision. However, although this module was developed to provide high-speed 3D compensation, the plastic balls in this experiment moved only in two dimensions. This is reasonable because a single camera can only provide information in two dimensions; however, this module has more potential.

1.4. Related Studies

To compensate for the 3D movements of the object, at least one additional sensor is required for the high-speed camera. Akamatsu et al. studied the 3D measurement of a moving object using a moving camera and a six-axis sensor [11], reducing the computational cost of 3D measurements compared to that using a stereo camera; however, the speed of this sensor was significantly lower than that of the high-speed camera used in the bead-art experiment.

Furthermore, there have been experiments and attempts at 3D real-time tracking, mostly using a single camera with an additional sensor [11, 12], or even without cameras [13]. Some of these methods only focus on tracking small-range movements, with the cameras kept stable [14] or moving in a small range [15]. Several approaches have tracked the movement of objects in the field of view, limiting the possible range of tracking.

The concept of 3D tracking and a robotic arm following the tracked object has been developed [16], but the operational speed was equivalent to that of humans [17], although the robotic system had the ability to move beyond humans because it did not have bodily limitations. Several similar experiments have attempted to predict the movement of an object, because some limitations in the camera frame rate cannot provide sufficient information. Grasping is a popular method in the field of 3D object tracking and manipulation [18, 19]. However, grasping generally involves only one-time contact between the robotic system and the object. When manipulation requires the system to continuously contact the tracked objects, grasping is insufficient. This means that the relative positions of the robotic system and the object need to be maintained within a small range.

More importantly, only a few of these studies focused on high-speed vision feedback. Higher-speed feedback can provide faster detection of movement changes and improve accuracy. The ability to process information and react at a frequency of 1000 Hz is beyond the processing ability of humans. The high-speed 3D compensation module developed in this study makes this possible.

1.5. Purpose

Previous studies have demonstrated the concept of real-time 3D object positioning and tracking using methods for

the 3D measurement of moving objects. The high-speed 3D compensation module has the potential to track objects in 3D movement with high-speed visual feedback, providing the potential for accurate real-time positioning in three-dimensions. Furthermore, the plastic balls in previous bead-art experiments were restricted to moving on a relatively small turntable, and a large range of possible movements in the assembly line of real industrial environments was not simulated. The experimental system provided the ability to move over a large range; however, this function was applied in bead-art assembly, from picking to building.

The purpose of this study is to accomplish high-speed 3D tracking of objects moving three-dimensionally over a relatively wide range using the high-speed 3D dynamic compensation module developed by our laboratory in collaboration with an industrial robot that moves with the object while maintaining its relative position. This system is also supposed to include a simple but valid high-speed 1000 Hz 3D visual measurement system with two moving cameras without the need for 3D calibration and complex considerations of their relative positions.

When completed, this system introduces the possibility of wide-range high-speed manipulation and manufacturing into the industry by adding manipulators onto the moving platform of the delta robot. This module can also be added to many existing industrial robots to upgrade their productivity with minor changes in the devices.

2. System Overview

The overall system comprised three types of devices. A real-world overview of the system is shown in **Fig. 2**. An overview of the entire system is shown in **Fig. 3**. The left side of **Fig. 3**, referred to as the operation devices, comprised three parts: a robotic arm (UR10e, Universal Robots), a high-speed dynamic compensation module, and two high-speed cameras that provide 3D visual feedback. The high-speed compensation module, later denoted as the delta robot because of its delta robot structure design, was assembled at the tool center of the robotic arm, with one high-speed camera assembled below the moving stage of the delta robot and vertically looking down. The structures of the UR10e and delta robots were similar to those assembled in a previous study. Delta robot's x - and y -axes were desired to be parallel to the robotic arm base system's x - and y -axes in this system. The delta robot's 3 degrees of freedom (3-DOF) feature made its moving stage move only in parallel, keeping this camera always facing vertically down. This camera (xy -camera) provides information about the horizontal plane, that is, the xy -plane of the robotic-arm base coordinate system. Another high-speed camera was attached to the base joint of the robotic arm and rotated along it. The z -camera provides information about the vertical plane and captures the z -axis movement of the tracked object. The two cameras were linked and synchronized to ensure that they sent the object status simultaneously. These comprised the actual "moving" devices in the

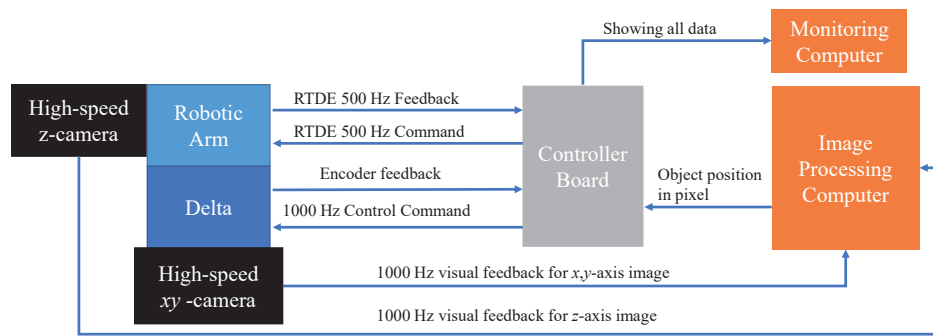


Fig. 3. An overview flowchart of the whole system.

3D tracking system. The middle device (referred to as the medium device) was the controller board. It can collect all the data from the separated devices, process them, and send the corresponding movement commands for 3D tracking to the robotic arm and delta robot. On the right-hand side, two computers comprised the section. The image processing computer was used solely for vision processing to ensure that the visual data remained at a speed of 1000 fps. Another computer monitored the controller board's data input and output during object tracking. It could also be used to write and revise programs on the controller board. UR10e provided the system with a wide range of movements, and the delta robot provided tracking accuracy and high-speed adaptability in terms of the reaction speed. Thus, their collaborative action achieved wide-range, high-speed 3D tracking.

The workflow of the system begins with the use of high-speed cameras that first collected image data at 1000 fps. Then, a vision-processing computer identified the position of the tracked object. The computer then sent the position of the object in pixels to the controller board. Based on these data, we calculated the position of the object in the robotic-arm base coordinate system and determined the location of the robotic arm. It sent the corresponding commands to move the robotic arm tool center to this position. However, because the maximum data translation speed for UR10e is only 500 times per second (500 Hz) and the tool speed and acceleration are limited, there is often a distance between the target position and UR10e's current position. In such cases, the controller board analyzes the error of the robotic arm using the position feedback of the robotic arm and vision data. It then sends commands to the delta robot to compensate for this error at a speed of 1000 Hz. As the devices moved, the cameras continuously provided images, and the image-processing computer updated the position information of the most recent object to the controller board. Consequently, the robotic arm and delta robot could be commanded together to achieve 1000 Hz object tracking.

3. Methodology

First, we define the names, notations, subscripts, and superscripts of the variables used in this section. Generally,

they follow the rules outlined below.

Uppercase X , Y , and Z in the main part of the notation indicate that the variable concerns the position in the direction of the robot base coordinate system. For example, X_{ob} is the x -position of the tracked object in the robot's base coordinate system.

Lowercase x , y , and z in the main part of the notation indicate that the variable concerns the position in the direction of the delta robot coordinate system. For example, x_{delta} is the current position of the delta robot in the delta robot coordinate system.

The lowercase d in the main part of the notation is typically used to represent the distance between various items. Uppercase D , meaning defined, is used for constants and can also appear on superscripts.

Superscripts and subscripts are used to describe positions or variables. For example, ob refers to objects, TCP is the tool center point, and xy indicates the xy -camera. One exception is when lowercase i is used to represent other variables or directions.

Each variable is explained in more detail when mentioned for the first time in the text.

3.1. High-Speed 3D Measurement System

In this section, f , c , u , and v represent the primary parts of the notation. They all concern the xy -camera and z -camera, and their superscripts and subscripts are shown.

3.1.1. Camera Position and Orientation

Generally, for a 3D measurement system, there are two cameras whose relative positions are stable, and the tracked object's position can be directly calculated after calibrating the two cameras together. However, this method is unsuitable for this system. For large-range tracking, at least one camera was required to move along with the robotic arm; otherwise, the robotic arm and other devices would become obstacles for tracking objects in images. Thus, it is not feasible to place two cameras outside to see the object, robotic arm, or delta robot together. However, the two cameras on a small stage of the delta robot also caused new problems. Although the camera was lightweight, a lens was required. When two cameras and two lenses were added to the stage of the delta robot, the reaction speed

decreased. Considering the possibility of adding a manipulator at this stage in the future, it would be preferable to decrease the load of the cameras on the stage as much as possible. More importantly, the tracking algorithm was expected to introduce oscillation to the tool center and the delta robot stage, which will cause errors in the camera. If two cameras are placed on the delta robot, the effect of this oscillation will be amplified. Considering these requirements, the positions and orientations of the cameras were determined, as shown in the system overview.

The xy -camera was placed under the stage of the delta robot, which was assembled at the tool center of the robotic arm. A z -camera was installed at the base joint of the robotic arm. It had the same rotation angle as that of the base joint, ensuring that it could always see the object moving in a large range and required a large angle change in the base joint of the robotic arm. The setting of the position and orientation of these two cameras was unusual; however, it solved the problems described previously and provided several important features for image processing. With this design, the robotic arm's and delta robot's degrees of freedom provide mobility to the camera.

3.1.2. Object Position Calculation

The RGB images from the cameras were converted to HSV images and binarized with HSV thresholds to detect an object that is supposed to have a unique color in the images. After the orientation and position of the cameras are adjusted through careful installation and adjustment using the results from the eye-in-hand calibration, the pixel data from the cameras can be directly used for calculating the position of the object. The X - and Y -positions of objects X_{ob} and Y_{ob} were first calculated as

$$X_{ob} = X_{TCP} + x_{delta} + \frac{u_{xy} - c_x^{xy}}{f_x^{xy}} d_{xy}, \quad (1)$$

$$Y_{ob} = Y_{TCP} + y_{delta} + \frac{v_{xy} - c_y^{xy}}{f_y^{xy}} d_{xy}, \quad (2)$$

with the encoder feedback from the robotic arm tool center position X_{TCP} and Y_{TCP} , the encoder feedback from the delta robot tool center position x_{delta} and y_{delta} , and xy -camera intrinsic parameters f_x^{xy} , f_y^{xy} , c_x^{xy} , and c_y^{xy} . u_{xy} and v_{xy} are the x - and y -pixel data of the xy -camera, respectively, and d_{xy} is the constant distance from the object to the xy -camera optical center. With these two positions and encoder feedback positions, the estimated distance from the object to z -camera optical center d^z is calculated, and using the same method, the z -position of the object is

$$Z_{ob} = Z_c^z + \frac{v_z - c_y^z}{f_y^z} \frac{d^z}{\sqrt{\left(\frac{u_z - c_x^z}{f_x^z}\right)^2 + \left(\frac{v_z - c_y^z}{f_y^z}\right)^2 + 1}}, \quad (3)$$

where Z_c^z is the z -position of z -camera optical center in the robot base system.

3.2. Robotic Arm Movement

After the position of the object was determined, the robotic arm tool center was commanded to be vertically above the object with a constant distance $d_{xy} + {}^{xy}D_{TCP}$, where ${}^{xy}D_{TCP}$ is the default vertical distance from the tool center to the xy -camera optical center when the delta robot was set to default position. Therefore, the command target position of the robotic arm was

$$[X_{ob}, Y_{ob}, Z_{ob} + d_{xy} + {}^{xy}D_{TCP}].$$

The movement command sent to the robotic arm was in the form of a speed-control command in three dimensions: x , y , and z . The command input was only for the tool center point (TCP) linear movement, as the orientation of the TCP was kept constant. Therefore, four degrees of freedom were used in the tracking method. For smooth movement involving possible severe direction changes and immediate target position updates, an algorithm for calculating the speed input was introduced.

Two numbers should be defined before the system shifts to the running mode. One is the maximum TCP linear moving speed in one axis, V_{max} , and the other one is the control gain k . First, when the target position was known, the first step was to calculate the distance from current TCP to it along each axis.

$$d_x = X_{ob} - X_{TCP}, \quad (4)$$

$$d_y = Y_{ob} - Y_{TCP}, \quad (5)$$

$$d_z = Z_{ob} + d_{xy} + {}^{xy}D_{TCP} - Z_{TCP}. \quad (6)$$

The control gain was then timed at this distance, and the actual speed input to the robotic arm is as follows:

$$\tilde{v}_i = V_{max} \left(-\frac{1 - e^{kd_i}}{1 + e^{kd_i}} \right), \quad (7)$$

where k is the control gain; i is equal to x , y , or z ; and $\tilde{v}_i$ and d_i are the input speed into the robotic arm tool center and the distance between the tool center and the target position on the i -axis, respectively. $\tilde{v}_x$ indicates that the TCP moves in the x -direction at a speed of $\tilde{v}_x$ [m/s]. $\tilde{v}_i$ are updated frequently for accurate 3D tracking.

The graph of $f(x) = -(1 - e^x)/(1 + e^x)$, as shown in Fig. 4, shows how this command provided smooth movement to the robotic arm. As $x \rightarrow 0$, $f(x) \rightarrow 0$ smoothly. Therefore, when the target position did not change, the change in status from approaching to stopping at the tool center was smooth. When the target position moved, the distance changed continuously, and the input speed changed smoothly and continuously on the graph.

By changing V_{max} , the overall input speed was scaled, and by changing k , the reaction speed was changed. As k increases, the speed changes more dramatically.

3.3. Delta Robot Movement

The intermediate variables used for the calculations in this section are labeled in the figures according to the following rules: lowercase b in the subscripts indicates the

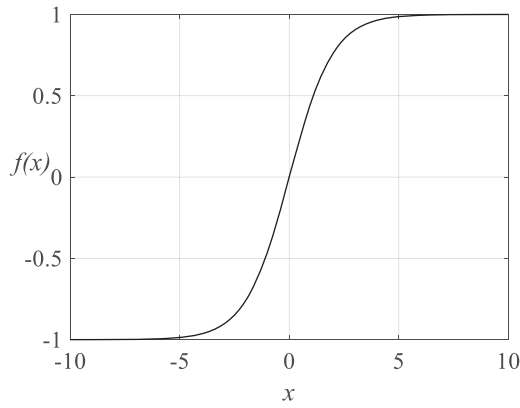


Fig. 4. Function for smooth movement: $f(x) = -(1 - e^x)/(1 + e^x)$.

ground base, where $Z = 0$ in the robot base coordinate system. The lower-case t in the superscripts indicates the target and dis is the abbreviation for displacement.

3.3.1. Delta Robot Displacement Required

The delta robot in this 3D tracking system was used to compensate for the real-time error between the robotic arm tool center current and target positions.

Before tracking started, the moving stage of the delta robot returned to its default position at $(0, 0, z_{delta}^D)$. Note that the origin $(0, 0, 0)$ was not at the TCP of the robotic arm. It was set at the center of the triangle formed by the three rotational axes of the three fixed motors. When tracking began, after the position of the tracked object was calculated as described in Section 3.1, and the robotic arm motion speed was calculated as described in Section 3.2, the target position of the delta robot needed to be determined.

The target position of the delta robot was calculated separately in the x -, y -, and z -coordinates. The x - and y -positions share similar settings. From the discussion in Section 3.2, it was observed that the robotic arm tool center was not able to directly “jump” to the target position as a step function. The movement was continuous and there was always a distance between the current and target positions. The delta robot was developed to compensate for this distance and achieve accurate real-time positioning. **Fig. 5** shows a diagram of the position relationship of the devices and tracked object in one tracking frame in the x -direction. At this moment, the target end position of the delta robot on the x -axis is

$$d_x = x_{delta} + x_{ob}, \quad (8)$$

where x_{delta} denotes the current end position of the delta robot. Because the fixed base of the delta robot is carried by the robotic arm, considering the input speed of the TCP, the displacement along x -axis for the delta robot at this moment for accurate positioning should be

$$\begin{aligned} x_{delta}^{dis} &= d_x - x_{delta} - \tilde{v}_x dt \\ &= x_{ob} - \tilde{v}_x dt, \end{aligned} \quad (9)$$

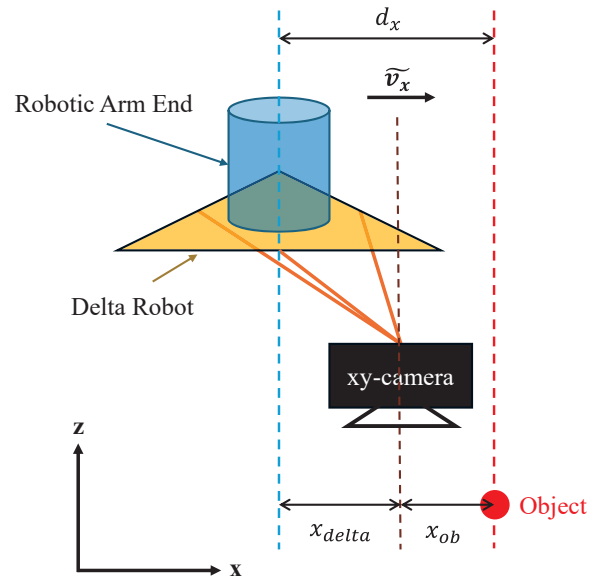


Fig. 5. Diagram of one frame about the x -axis during tracking.

where dt is 1 ms and the time between the two commands sent to the delta robot.

Because the movements in x - and y -directions had the same setting, the displacement of the delta robot in y -axis was

$$y_{delta}^{dis} = y_{ob} - \tilde{v}_y dt. \quad (10)$$

However, as shown in **Fig. 6**, in the z -direction, the calculation was different. The target position in z -direction of the robotic arm tool center was determined previously, and the distance between the current TCP and target TCP d_z was also calculated as described in Section 3.2. At this moment, the target end position of the delta robot along the z -axis of the robot base system should be

$${}^b z_{delta}^t = Z_{ob} + d_{xy} + {}^c D_{TCP} + z_{do}^D + z_{delta}^D, \quad (11)$$

where z_{do}^D is the vertical distance from the origin of the delta robot to the TCP. Note that z_{do}^D , z_{delta}^D , and z_{delta} are typically negative because they are displaced in the negative z -direction of the base system. The current position of the delta robot in the same system was

$${}^b z_{delta} = Z_{TCP} + z_{do}^D + z_{delta}. \quad (12)$$

Therefore, the target end position for the delta robot in the z -direction in its own system should be

$$\begin{aligned} z_{delta}^t &= {}^b z_{delta}^t - {}^b z_{delta} \\ &= Z_{ob} + d_{xy} + {}^c D_{TCP} + z_{do}^D + z_{delta}^D \\ &\quad - (Z_{TCP} + z_{do}^D + z_{delta}) \\ &= d_z + z_{delta}^D - z_{delta}, \end{aligned} \quad (13)$$

with

$$d_z = Z_{ob} + d_{xy} + {}^c D_{TCP} - Z_{TCP}.$$

The displacement of the delta robot in the z -direction was

$$\begin{aligned} z_{delta}^{dis} &= z_{delta}^t + z_{delta}^D - z_{delta} - \tilde{v}_z dt \\ &= d_z + 2z_{delta}^D - 2z_{delta} - \tilde{v}_z dt. \end{aligned} \quad (14)$$

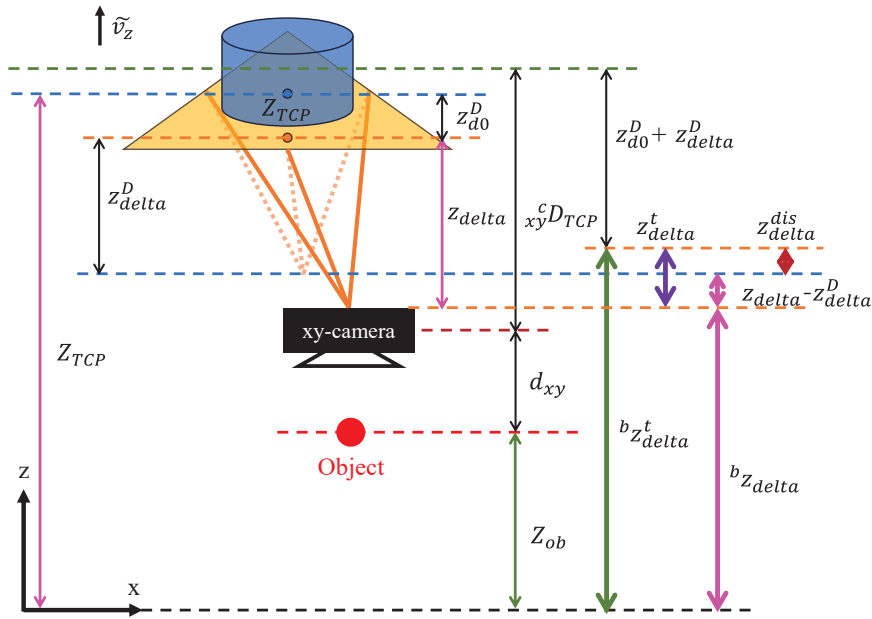


Fig. 6. Diagram of one frame about the z-axis during tracking.

3.3.2. Delta Robot Movement Control

When controlling the movement of the robotic module by implementing the delta structure at a speed equivalent to that of the high-speed 3D compensation module, the displacement in each cycle of commands was regarded as the speed, because each cycle finished in approximately 1 ms, and each commanded displacement could be maintained in several cycles. Therefore, using the linear displacements in the x-, y-, and z-directions, x_{delta}^{dis} , y_{delta}^{dis} , and z_{delta}^{dis} were calculated as described in Section 3.3.1. The corresponding required angular displacements of motors in the delta robot, θ_{dis}^1 , θ_{dis}^2 , and θ_{dis}^3 , were calculated using the kinematic equations of the delta robot explaining the relationship between the input speed in x-, y-, and z-directions and the corresponding required rotation speed of the motors [a].

To better control the reaction speed and movements of the delta robot, a gain K_d was added to the calculated required linear displacements from Section 3.3.1 when calculating the angular displacements. The final angular displacement equation is as follows:

$$K_d \begin{bmatrix} x & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} x_{delta}^{dis} \\ y_{delta}^{dis} \\ z_{delta}^{dis} \end{bmatrix} = \begin{bmatrix} B_{11} & 0 & 0 \\ 0 & B_{22} & 0 \\ 0 & 0 & B_{33} \end{bmatrix} \begin{bmatrix} \theta_{dis}^1 \\ \theta_{dis}^2 \\ \theta_{dis}^3 \end{bmatrix}, \quad (15)$$

where

$$\begin{aligned} A_{21} &= 2(x+b) - \sqrt{3}L \cos \theta^2, \\ A_{31} &= 2(x-b) - \sqrt{3}L \cos \theta^3, \\ A_{12} &= y+a + L \cos \theta^1, \\ A_{22} &= 2(y+c) - L \cos \theta^2, \\ A_{32} &= 2(y+c) - L \cos \theta^3, \end{aligned}$$

$$A_{13} = z + L \sin \theta^3,$$

$$A_{23} = 2(z + L \sin \theta^3),$$

$$A_{33} = 2(z + L \sin \theta^3),$$

$$B_{11} = L[(y+a) \sin \theta^1 - z \cos \theta^1],$$

$$B_{22} = -L \left[\left(\sqrt{3}(x+b) + y + c \right) \sin \theta^2 - 2z \cos \theta^2 \right],$$

$$B_{33} = L \left[\left(\sqrt{3}(x-b) - y - c \right) \sin \theta^3 - 2z \cos \theta^3 \right],$$

and

$$a = m_B - n_P,$$

$$b = \frac{S_P}{2} - \frac{\sqrt{3}m_B}{2},$$

$$c = m_P - \frac{m_B}{2}.$$

Here, x , y , and z are the current delta robot positions in the three directions. m_B , n_P , S_P , and m_P are shown in Fig. 7. L is the motor arm length from Fig. 1, and θ^1 , θ^2 , and θ^3 are the current angles of the motors as defined in Fig. 1.

After solving θ_{dis}^1 , θ_{dis}^2 , and θ_{dis}^3 , a controller was added to output the required torque τ^i for the corresponding motors with the form:

$$\tau^i = k_g \left[k_p (\theta_p^i + \theta_{dis}^i - \theta^i) - k_d \frac{\theta^i - \theta_p^i}{T_s} \right], \quad (16)$$

where $i = 1, 2$, or 3 , representing the corresponding motors. θ_p^i is the target angle from the last cycle, θ^i is the current angle from the encoder, θ_p^i is the angle recorded by encoder in the last cycle. k_g , k_p , and k_d are the controller gains. T_s is the sampling time. The calculated τ^i was limited from -1 to 1 and then transformed to voltage input to the motors.

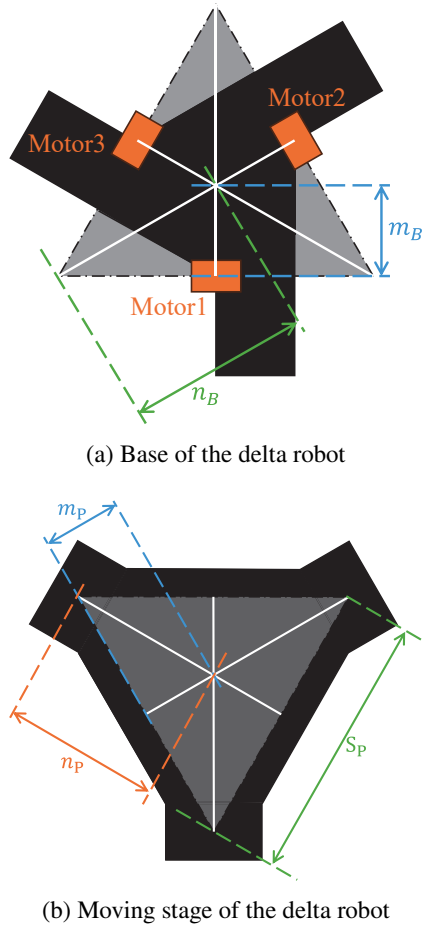


Fig. 7. Diagrams of the base and moving stages of the delta robot shown in **Fig. 1**.

When the delta robot was running rapidly, it was assumed that θ_p^i could be achieved in one cycle; then $\theta_p^i = \theta^i$, and the equation was simplified to

$$\tau^i = k_g \left[k_p \theta_{dis}^i - k_d \frac{\theta^i - \theta_p^i}{T_s} \right]. \quad (17)$$

4. Evaluation of the Developed Wide Range High-Speed 3D Tracking System

4.1. Tracked Object

In this 3D tracking system, the tracked object was designed to move along a relatively stable path, including moving up and down and turning left or right in a circuit for better comparative evaluations.

Considering the size of the object in the cameras, it was finalized as a red ball with a diameter of 19 mm, which is a standard marker. A red ball was installed on a toy locomotive running on a plastic railway circuit. This circuit was formed as a circuit full of up and down movements to evaluate the entire tracking system. The movements in this circuit can be roughly arranged into four types: downhill straight, downhill U-turn, uphill straight, and uphill

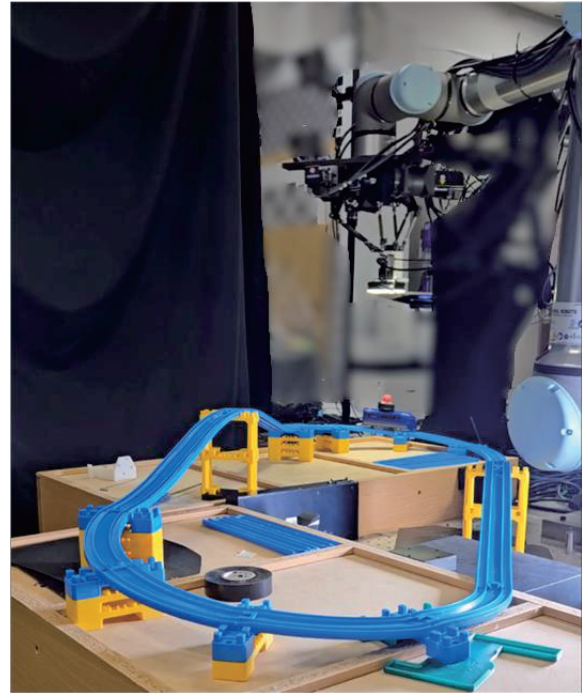


Fig. 8. Experimental setup.

U-turn. As they were not precise-scale models, the toy locomotive ran within tolerance, leading to system uncertainties. For each experiment, the starting point of the locomotive was set to the highest point of the circuit that provided the highest acceleration at the beginning. **Fig. 8** shows a photograph of the toy locomotive running on the circuit when the 3D tracking system was on.

4.2. Evaluation of Robotic Arm Control Algorithms

First, the tracking experiment with only UR10e was powered on, and the delta robot was maintained at the default position. The gain k of the robot in Eq. (7) was set to 3.14, and $V_{\max}$ was set to 5 m/s. The system was run for 40 s, and there was no robotic action in 0–5 s, and 35–40 s was used for returning to the starting point. Thus, the tracking was run between 5 s and 35 s. **Fig. 9** shows the location of object in the cameras at an instant time during tracking, and **Fig. 10** plots the 3D position of the object for the entire tracking period. Movement did not start immediately after tracking started; therefore, the operator could ensure that everything in the system was running correctly.

From **Fig. 9**, it can be observed that when only UR10e was tracking, the red ball was not always at the center of the xy -camera (the intersection of the white cross), which meant that this TCP speed input was not able to capture the moving object perfectly. This is slightly slower than that of the object. The red and blue lines in **Figs. 10(a)** and **(b)** did not overlap (part of the graph was enlarged to show this), which also proved this. The errors in the TCP positions and the object positions were considered as the “uncertainty” from the robotic arm, as discussed in Section 1. The difference between the two graphs is not large, with

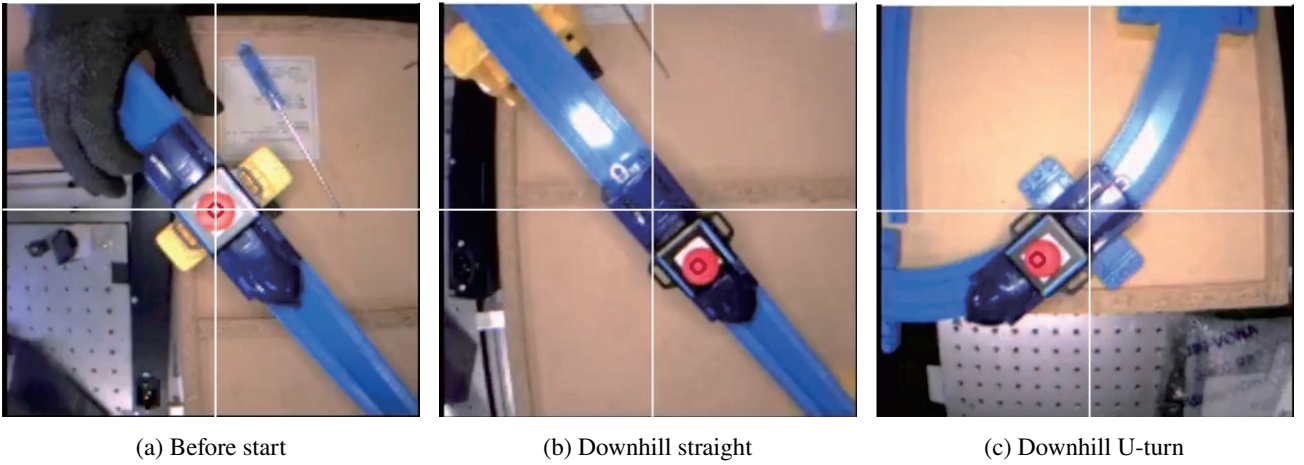


Fig. 9. Object in xy -camera in different parts of the circuit when only robotic arm was active.

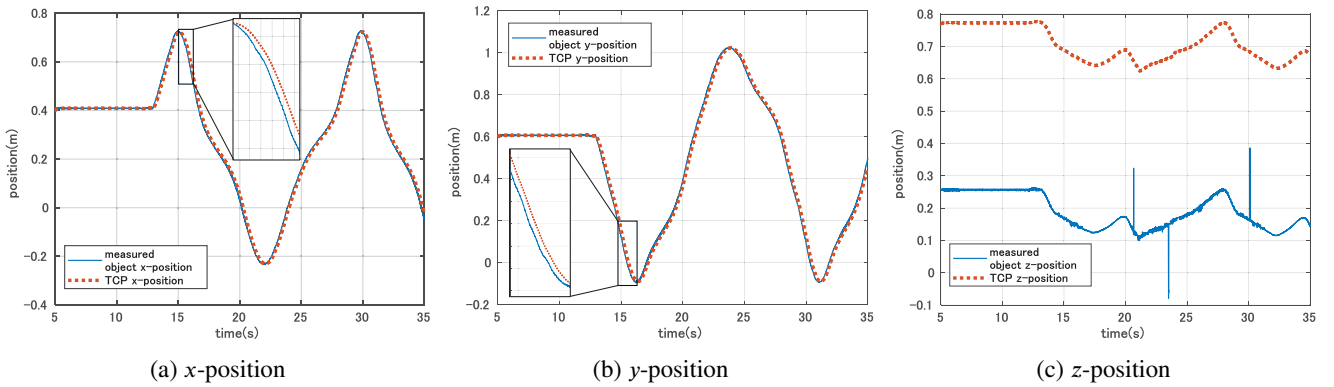


Fig. 10. TCP and object position in the x -, y -, and z -axes when only robotic arm was active.

an average tracking error of 1.89 mm on the x -position, and 0.50 mm on the y -position. In general, a robotic-arm control-tracking algorithm is feasible.

Because of the relatively small change in the z -axis, tracking in the z -position was better when only the UR was on. As shown in **Fig. 10(c)**, although some outliers (sudden peaks of data) were observed. Because the camera was fixed to the robot arm for the z -position measurement, the moving target could be far away from the camera resulting in outliers. The z -axis tracking was considered successful because the robot was controlled at a high frame rate, as shown in **Fig. 11**. The average tracking error along z -axis was 0.66 mm.

4.3. Evaluation of Tracking with Collaboration of Robotic Arm and Delta Robot

After tracking using only UR10e, the high-speed 3D compensation module, the delta robot, was added into experiment to improve the tracking, trying to fix the “uncertainties” involved during only UR10e tracking.

The controller gains (defined in Section 3.3.2) of the delta robot are set as follows:

$$K_d = 0.5, \quad k_g = 1, \quad k_p = 13, \quad k_d = 18.$$

The object in the cameras is shown in **Fig. 12**, in comparison with a similar moment when only the robotic arm

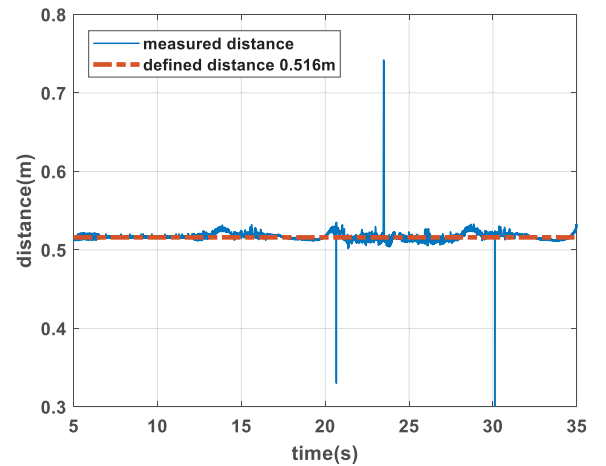


Fig. 11. Measured distance between the object z -position and TCP z -position, with their defined distance 0.516 m.

was active. The red ball was closer to the center of images of xy -camera, although the detected center was not always at the center of the red ball. The 3D positions are plotted in **Fig. 13**, and this time from the enlarged part, it is clear that the red and blue graphs overlapped, which means that the x - and y -axis tracking became successful after the involvement of the delta robot. The average tracking error is 0.06 mm on x -position, and 0.09 mm on y -position. The

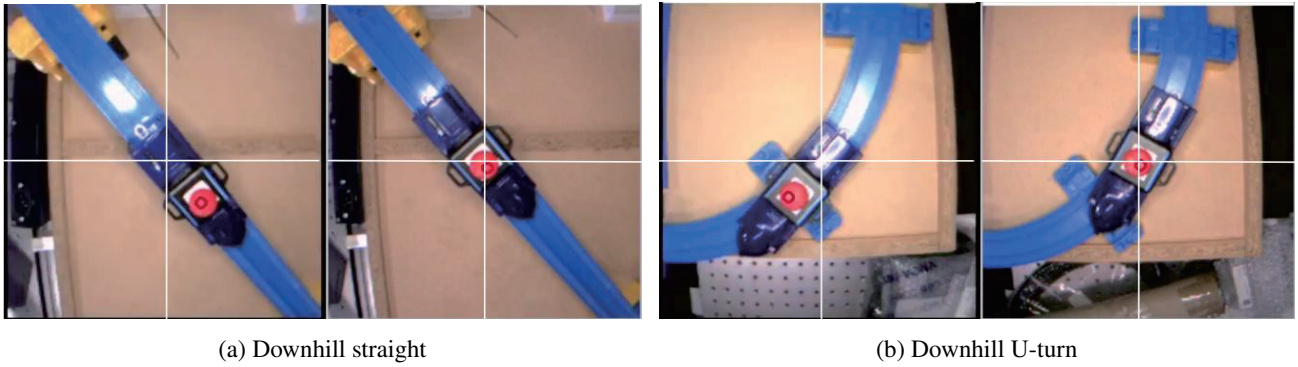


Fig. 12. Comparison of object in xy -camera in different parts of the circuit between when robotic arm was active (left) and when robotic arm and delta robot were active (right).

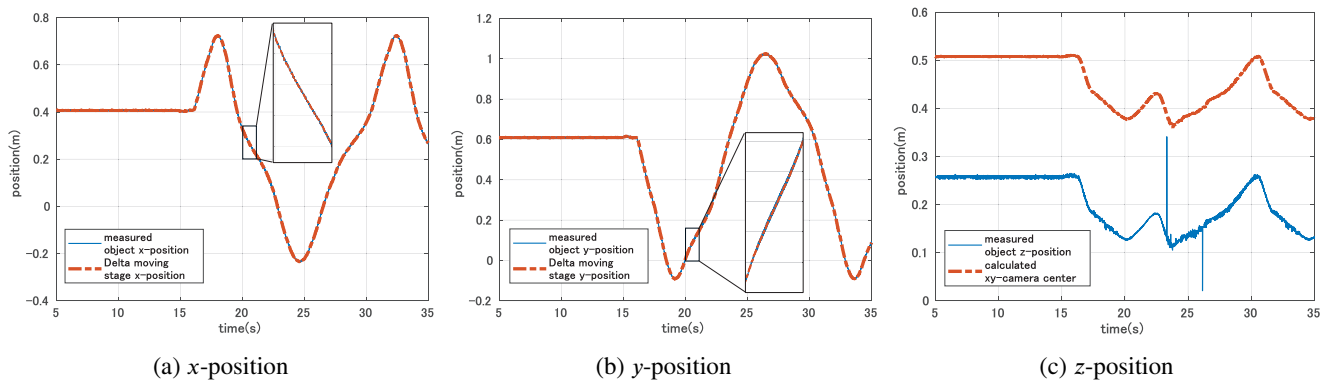


Fig. 13. TCP and object position in x -, y -, and z -axes when robotic arm and delta robot were active.

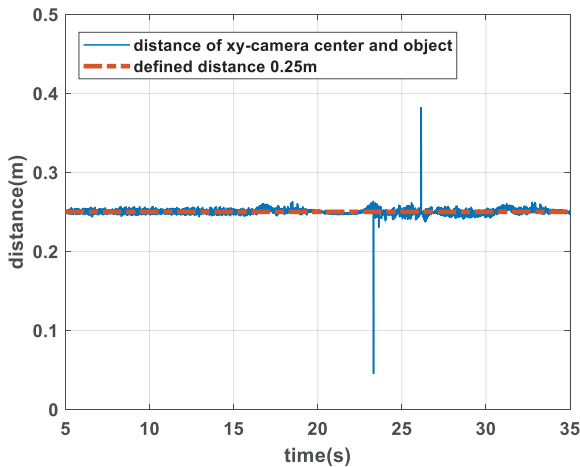


Fig. 14. Measured distance between the xy -camera center and the object, with their defined distance 0.25 m.

vertical distance between the object and xy -camera was defined to be 0.25 m, and the average tracking error in z -axis decreased to 0.2 mm, as shown in Fig. 14. Therefore, the tracking algorithm and this collaboration for 3D tracking are proved feasible.

4.4. Evaluation of the 3D Tracking System Tracking High-Speed Moving Object

To further evaluate the resilience of the 3D tracking system, the chassis of the toy locomotive was exchanged for

a high-speed chassis, and the same process as before was performed with the gain k in Eq. (7) set to 6.28 for the high-speed reaction from UR10e. The average speed of the chassis was approximately 0.33 m/s, and in the fastest downhill straight section, it increased to approximately 0.7 m/s. The tracking data are shown in Fig. 15.

A higher speed can be proven by comparing Figs. 13 and 15, focusing on the number of cycles that the object ran. In this experiment, x - and y -axis tracking was successful. The average tracking error is 0.02 mm on the x -position, and 0.19 mm on the y -position. The average error in z -axis tracking from 5 s to 35 s was approximately 0.1 mm, as shown in Fig. 16.

4.5. Discussion

Three experiments are conducted to evaluate the performance of the 3D tracking algorithm. The results of the three experiments are listed in Table 1.

The evaluation of this wide range of 3D tracking with several different experimental settings proved the feasibility of using a high-speed 3D compensation module and robotic arm together to achieve wide-range 3D tracking by comparing the average tracking error with and without the delta robot. However, in almost all tracking data, there were still noticeable vibrations at the moving stage, which might prevent further devices from being added to the stage for more functions of this system.

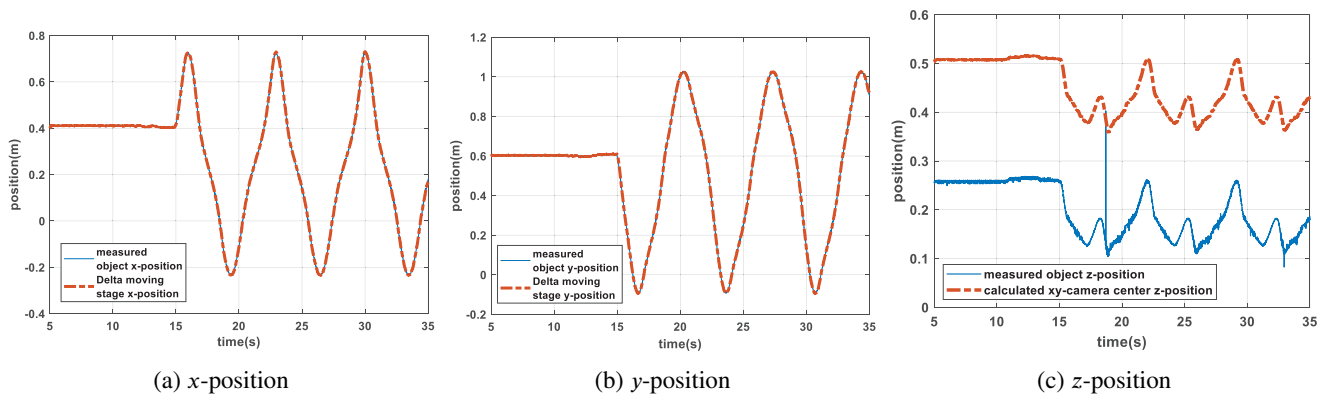


Fig. 15. TCP and high-speed moving object position in x-, y-, and z-axes when robotic arm and delta robot were active.

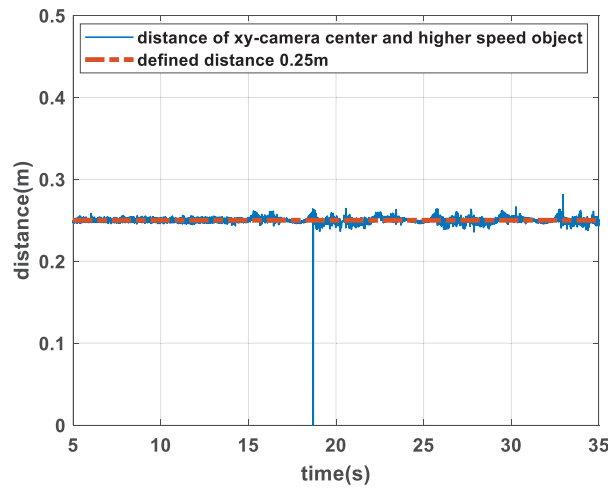


Fig. 16. Measured distance between the xy-camera center and the object with higher speed, with their defined distance 0.25 m.

Table 1. Brief summary on the performance of 3D tracking experiments.

Experiment setting	x-axis tracking evaluation	y-axis tracking evaluation	z-axis tracking evaluation	x-axis average tracking error	y-axis average tracking error	z-axis average tracking error	Overall tracking result
Only robotic arm	Incomplete	Incomplete	Successful	1.89 mm	0.50 mm	0.66 mm	Failure
Collaboration of robotic arm and delta	Successful	Successful	Successful	0.06 mm	0.09 mm	0.20 mm	Success
Collaboration tracking an object with higher speed	Successful	Successful	Successful	0.02 mm	0.19 mm	0.10 mm	Success

5. Conclusion

In this study, a wide-range 3D tracking system was developed using a robotic arm and a high-speed 3D compensation module. This system can track an object moving over a wide range while the end position of the combined robotic devices is always at the same distance above the object. This tracking system used two high-speed cameras to provide 1000 Hz visual feedback, and the high-speed compensation module provided 1000 Hz responsibility for movement with high accuracy. Especially in the x-axis, the tracking error was less than 0.1 mm. The robotic arm achieves this accuracy over a wide range. The tracking accuracy is higher when the object moves faster. This

study showed the feasibility of using a high-speed 3D dynamic compensation module for wide-range object tracking, which may contribute to the higher-speed manipulation of robotic arms to increase productivity.

To further improve the system, additional controller methods and combinations of current parameters should be tested to decrease vibration. After solving the vibration problems, high-speed 3D manipulation can be realized.

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